

CTS SRE Day 8

Observability and Microservices

EMPOWERING High Performance
Technology Teams

Agenda

Duration	Activity
10 mins	Welcome and Objectives Implement end-to-end observability for microservices application in AWS.
10 mins	Observability Tools & Implementation
10 mins	Lab Overview – Observability for Microservices Gain hands-on experience with monitoring, logging, and tracing tools.
10 mins	Labs Steps Overview Configure observability tools for metrics collection with custom exporters & service discovery.
10 mins	Break
80 mins	Hands-On Labs – Step-by-Step Implementing Prometheus, NodeExporter, Grafana Setup and Dashboard Metric Creation, NodeJs Application for Banking App, AWS Instance
10 mins	Summary
10 mins	Q&A

Observability

- **Definition:** Observability is the ability to measure a system's internal state based on its external outputs.
- **Importance in Observability:**
 - Helps identify issues in distributed microservices.
 - Enables proactive incident response
 - Improves system reliability

- **Three Pillars of Observability:**

Logs – Capturing system events

Metrics – Measuring system performance (CPU, Latency, Error Rate)

Traces – Tracking requests across microservices.

Observability Tools & Implementation

- **Tools for Observability:**

Logging: ELK Stack, Fluentd

Metrics: Prometheus, Grafana, AWS CloudWatch

Tracing: Jaeger, OpenTelemetry, AWS X-Ray

- **Best Practices:**

Use structured logging for better analysis

Monitor key metrics (CPU, latency, error rate)

Implement distributed tracing for debugging

Automate alerts & incident response for SRE efficiency

Lab Overview – Observability for Microservices

◆ What Participants Will Learn

Implement a comprehensive observability setup for microservices application.
Deploy and configure monitoring, logging, and tracing tools.
Gain hands-on experience with Prometheus, Node Exporter and Grafana.

◆ Key Learning Areas

Metrics Collection: Configure **Prometheus** for collecting microservices performance data.

Real-time Visualization: Set up **Grafana dashboards** for latency, error rates, and KPIs.

Hands-On Labs – Step-by-Step

◆ Lab Activities

- **Set Up Prometheus and Node Exporter:**

Configure Prometheus for **metrics collection** in a microservices architecture.

Implement **custom exporters** and **service discovery**.

Define **recording rules** for key performance indicators.

- **Deploy Grafana Dashboards:**

Set up **Grafana** for real-time monitoring.

Create dashboards for **latency, error rates, throughput, and resource utilization**.

- **Monitor a Banking Microservices Application:**

Apply Prometheus monitoring to track system health.

Visualize business KPIs for a Banking use case.

Lab Guides

- Available on the LMS under (lab Content)
- Github Link: https://github.com/AshishSharmaPrivate/Lab/blob/main/Observability_for_Microservices_Applications.txt

Summary & Key Takeaways

- Observability Implementation of End-to-End Microservice Application.
- Dashboard creation for various metrics (CPU, Memory, IO etc) on Grafana.
- Visualizing Banking App metrics (Payment, Balance etc).

Thank You!
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